INCH-POUND

MIL-T-50320C(AR)
18 May 1990
SUPERSEDING
MIL-T-50320B(MU)
20 December 1974

MILITARY SPECIFICATION

TRANSMIT GATE: 10549202

This specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a Transmit Gate used in Receiver-Transmitter RT-860, part of Radar Set ANIVPS-2 and in Subassembly Test Set MX-8048,part of Radar Test Set AN/TPM-22.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbocks. The following specifications, standards, and handbooks form a part' of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-F-13926 - Fire Control Materiel, General Specification Governing the Manufacture and Inspection of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander U.S. Amy ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC1285

DISTRIBUTION STATEMENT A Approved for public release; distribution is unlimited.
Source: https://assist.dla.mil - Downloaded: 2026Jul-22T20:21Z
Check the source to verify that this is the current version before use.

MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-T-50671	Transmit Gate, Board B: 10549207
MIL-T-50672	Transmit Gate, Board A: 10549205

STANDARDS

MILITARY

MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods 'and Engineering Guidelines
MIL-STD-2000 -	standard Requirements for Soldering Electrical and Electronic Assemblies

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS (see 6.4)

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

10549202 - Transmit Gate

PACKAGING DATA SHEET

PS10549202 - packaging of Transmit Gate

(Copies of drawiings, publications, and other Government documents required by centractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of conflict between the text of this specification and the references cited herein (except for associated-detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:21Z
Check the source to verify that this is the current version before use.

3. "REQUIREMENTS

3.1 **Materials.** Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 10549202, all associated drawings, and with all requirements specified in applicable specifications and standards. The subassemblies of the transmit gate shall have been inspected and accepted *in* accordance with the drawing prior to assembly in the transmit gate. The following subassemblies are covered by the indicated specifications:

Subassembly		Specification
Transmit Gate, Board A:	10549205	MIL-T-50672
Transmit Gate, Board B:	10549207	MIL-T-50671

Where applicable, the transmit gate shall meet the requirements of MIL-F-13926 as follows:

a. Dimensions and tolerances
b. Inorganic protective surface finishings
C . part identification and marking
d. Workmanship

3.3 **Environmental service conditions.**

3.3.1 Vibration. The transmit gate shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to sinusoidal vibrations, in each of three mutually perpendicular axes, as follows:

Frequency Range	Amplitude
10 to 15 hertz	0.06 inches double amplitude
15 to 25 hertz	0.04 inches double amplitude
25 to 55 hertz	O.O2 inches double amplitude

The frequency of vibration shall be swept logarithmically.
The cycling time shall be 120 minutes per axis.
The sweep rate shall be 3.75 minutes ± 10% per decade.
The 10-55-10 hertz sweep time is 3 minutes 45 seconds ± 10%.

3.3.2 **Shock.** The transmit gate shall not suffer damage and shall meet requirements of 3.4 and 3.5 after exposure to a total of 18 impulse shocks consisting of three shock impulses of 15 ± 10% g's, 11 ± 10% milliseconds, half-sine wave applied in each direction of 3 mutually perpendicular axes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:21Z
Check the source to verify that this is the current version before use.

3.3.3 Altitude.

3.3.3.1 Operating. The transmit gate shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches of Hg).

3.3.3.2 Non-operating. The transmit gate shall meet the requirements of 3.4 and 3.5 after exposure for not less than 10 minutes to barometric pressure encountered at 40,000 feet above sea level (5.5 inches of Hg).

3.3.4 Temperature.

3.3.4.1 Operating. The transmit gate shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between -40 degrees F and +155 degrees F.

3.3.4.2 Non-operating (storage). The transmit gate shall not suffer permanent damage when subjected to the constant temperature extremes of -60 degrees F for a period of 24 hours and +160 degrees F for a period of 4 hours (after temperature stabilization) and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity (non-operating) The transmit gate shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.4 Electrical. Each conductor shall be electrically continuous as specified on the schematic diagram of the transmit gate.

3.5 Performance. When inputs are applied to the transmit gate as described below, the outputs shall be as stated below.

3.5.1 Input voltages. The transmit gate shall be powered by the following voltages:

+15 ± 0.1 Vdc between J4-1(+) and J4-3 (Ret)

-15 ± 0.1 VDC between J4-5(-) and J4-3 (Ret)

3.5.2 Modulator trigger. When a modulation trigger pulse train with a PRE of 20 ± 1 KHz, pulse width of 0.5 ± .1 us, amplitude of +10V minimum, and a rise time of less than 100 ns is applied to J1 and power is applied per 3.5.1, the following outputs shall be observed at J2:

a. A DC level which is adjustable to within 0 ± .002vdc of ground using R24, Transmit Gate Mod D.C. Offset adjustment, if necessary. See Figure 1.

Source: https://assist.dla.mil - Downloaded: 2026-10-22T20:21Z
Check the source to verify that this is the current version before use.

b. The above output shall be adjusted for maximum waveform flatness using R31, Transmit Gate D.C. Offset adjustment. See Figure 2.

C. An output shall be present at J6 which is identical to the input pulse applied to J1.

3.5.3 Rf signal. With conditions as stated in 3.5.2 and a 60 ± 1 MHz, 2.3 ± IV RMS (+20 dBm) Rf signal applied to J3, the output observed at J2 shall be an RF burst of cosine square shape, with a burst repetition rate equal to that of the modulation trigger pulse and an RF frequency equal to the RF input, and amplitude of 1V minimum peak-to-peak, and a burst width of 280 ± 10 nanoseconds between the 50% amplitude points (see Figure 3).

3.5.4 Spectral purity. The first side lobe of the output spectrum shall have a maximum amplitude of not greater than -26dB with respect to the center lobe. The second side lobe shall have a maximum amplitude of not greater than -35dB with respect to the center lobe (see Figure 4).

3.5.5 Rf signal leakage. With conditions as in 3.5.3 and the modulation trigger input removed from Jl, the measured peak-to-peak leakage at J2 shall be no greater than that calculated using the following formula:

$$E_{P-P} = \frac{E_{P-P} \text{ (measured amplitude of 3.5.3)}}{70.79}$$

3.6 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.7 Workmanship. The workmanship requirements of MIL-F-13926 shall apply.

3.8 Soldering. All soldering shall be in accordance with the general requirements of MIL-STD-2000 in addition to detail requirement (task)F of paragraph 1.2 of MIL-STD-2000.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:21Z
Check the source to verify that this is the current version before use.

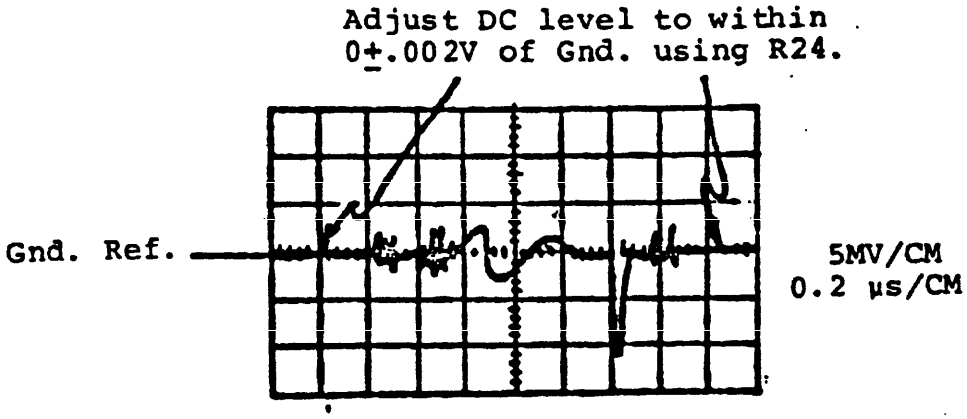


FIGURE 1. R24 adjustment.

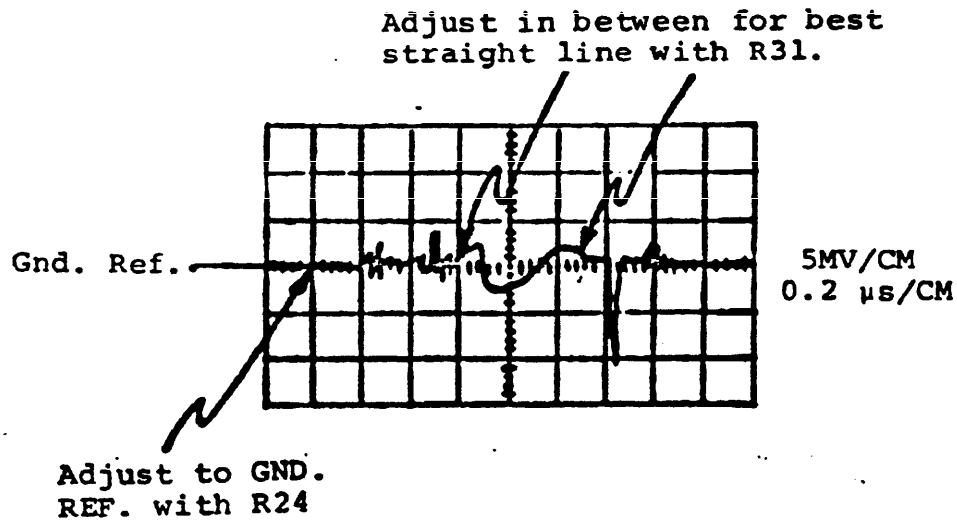


FIGURE 2. R31 adjustment.

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:21Z
Check the source to verify that this is the current version before use.

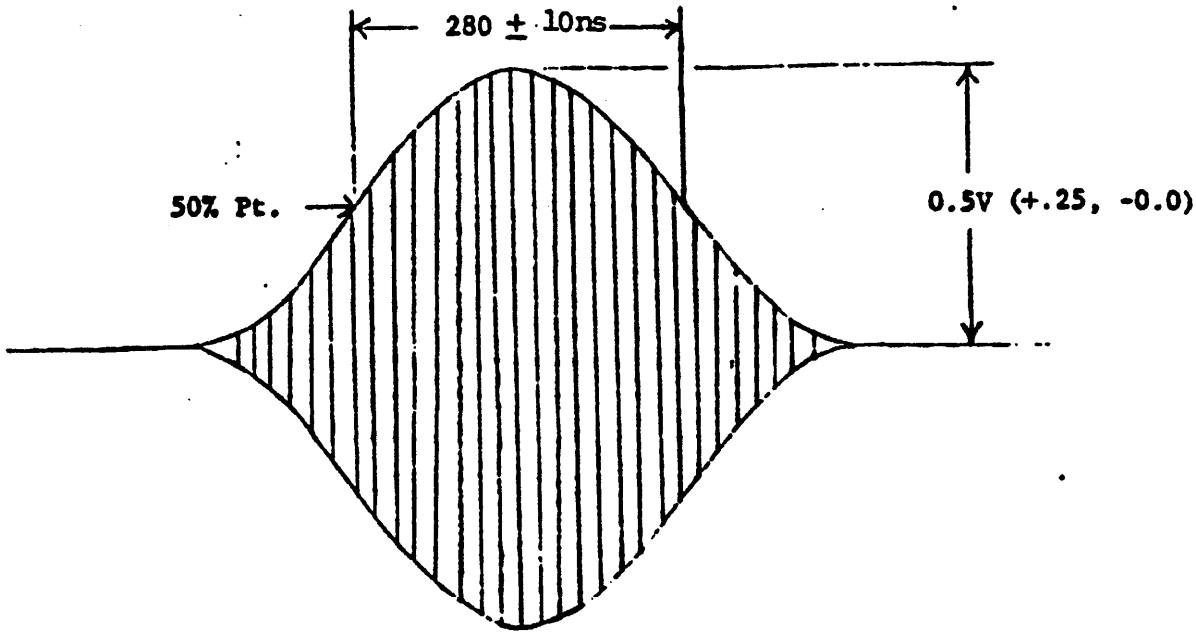


FIGURE 3. Rf burst.

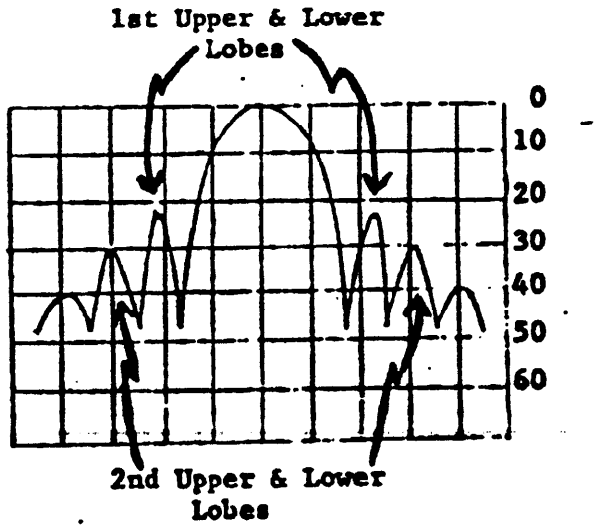


FIGURE 4. Output spectrum.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:21Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may *use* his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.1.1 **Responsibility for compliance.** All items must meet all requirements of sections 3 and 5 and all applicable drawings. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Special sampling inspection (see 4.5).

4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a.	Temperature	+50 deg to +90 deg F.
b.	Relative humidity	90% RH Max.
c.	Pressure	28 to 32 inches of Hg.

4.3 **First article inspection.**

4.3.1 **First article submission.** The requirement for first article and the responsibility (Government or contractor) for the first article testing shall be as specified in the contract and the applicable drawing. The sample for the first article tests shall consist of three transmit gates manufactured in the same manner,

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:21Z
Check the source to verify that this is the current version before use.

using the same materials, equipment, processes and procedure as in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.3.1.1 Government testing. When the Government is responsible for conducting first article tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to ensure that it conforms to all the requirements of the contract and the applicable drawings. The contractor shall submit a record of this inspection with the sample including certified test report (CTR) or certificate of conformance (cOC) for the materials.

4.3.1.2 Contractor testing. When the Contractor is responsible for conducting first article tests, the sample shall be inspected by the contractor for all the requirements of the contract, all of the examinations and tests specified in Table 1 (Functional Test) and Table II (Special Sampling) herein and to all requirements of the applicable drawings. The sample and a record of this inspection, including CTR/COC, shall be submitted for the Government approval. The Government reserves the right to witness the Contractor's inspection.

4.3.2 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any assembly. component or test specimen to comply with any of the stated requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term 'inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, In accordance with the same drawings, same drawing revision, same specificaticas and same Specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examinations and tests.

4.4.2.1 Components, assemblies and subassemblies. All components, assemblies and subassemblies shall be inspected in

Source: https://assist.dla.mil -- Downloaded: 2026l0-22T20:21Z
Check the source to verify that this is the currentversion before use.

accordance with the inspection provisions contained in this specification and the Quality Assurance Provisions (QAP) listed in the Technical Data Package (TDP). In the absence of QAP's, the applicable quality assurance provisions of MIL-F-13926 shall apply.

4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with the sampling plans specified in 4.4.2.3 herein. Examination and test for packaging and marking shall be in accordance with Section 5 herein. The tabulated classification of defects in 4.4.2.3 shall constitute the minimum inspection to be performed by the supplier after first article aproval and prior to Government acceptance or rejection by item or lot.

4.4.2.3 Functional tests. The tests in Table I shall be performed on a 100% basis.

TABLE 1. Functional tests
(100% Inspection)

CHARACTERISTIC		REQUIREMENT	TEST PROCEDURE
Critical:		None	N/A
MAJOR			
101.	Electrical	3.4	*4.7.2*
102.	Performance	3.5	*4.7.3*
Minor:		None	N/A

4.5 Special sampling. One transmit gate out of the first 50 produced and one out of each 100 produced thereafter shall be selected at random. The sample shall have met the requirements and tests of Table I and shall then meet the tests in Table II. Unless otherwise specified in the contract the items shall be returned to the lot upon successful completion of testing.

TABLE 11. Special sampling

CHARACTERISTIC		REQUIREMENT	TEST PROCEDURE
Critical:		None	N/A
Major:			
*103**	Vibration	3.3.1	4.7.1.1
104.	Shock	3.3.2	4.7.1.2
105.	Altitude	3*3.3	4.7.1.3
106.	Temperature	3.3*4	4.7.1.4
107.	Humidity	3.3.5	4.7.1.5

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:21Z
Check the source to verify that this is the current version before use.

Minor:	None	N/A

4.5*1 Failure of sample. Should any one item of a special sampling fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test as well as to current production has been accomplished in accordance With the provisions of the contract.

4.6 Inspection equipment.

4.6.1 Government-furnished inspection. equipment. Where the contract provides for Government-furnished test equipment Supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.6.2 contractor-furnished inspection equipment

4.6.2.1 Government design. Unless otherwise specified in the contract all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The centractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Centractor design. The contractor shall design and supply inspection equipment compatible with the "Methods of Inspection" specified in 4.7 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926 whenever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Cconstruction shall be such as to facllitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordante with 6.3.

4.6.2.3 Test equipment. In conjunction with 4.6.2.1 and *4.6.2*.2, equipment listed in Table III shall be used in the performance of the applicable test methods and procedures as specified in 4.7.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:21Z
Check the source to verify that this is the current version before use.

TABLE III. Test equipment list

ITEM NUMBER	DESCRIPTION
1. power supply	+15 ± .1 Vdc, ac ripple less than 50 mV peak-to-peak; capacity 100 mA.
2. Power supply	-15 ± .1 Vdc, ac ripple less than 50 mv peak-to-peak; capacity 100 MA.
3. Oscillator	Capable of being set to and delivering 60 ± 1 MHZ, amplitude 2.3 ± .lV RMS, into a 50 ohm load.
4. Pulse generator	capable of being set to and delivering PRF 20 ± 1 KHz, width of .5 ± .1 us, amplitude of +10V, rise time less than 100 ns into a 50 ohm load.
5. Oscilloscope	Dual channel. Vertical passband dc to 150 MHz . Rise time of scope plus measuring probe 3.5 ns or less. Sweep period 5O ns per division or less. DC sensitivity Of 5 mV/cm. Input resistance greater than 1 megohm. Input capacitance less than 25 pF.
6. Test harness	See Figure 5. See Table IV for parts list.
7. Spectrum analyzer	Frequency range 10 MHz to 1 GHz minimum. Capable of displaying a 30-MHZ spectrum width with a 60-dB dynamic range and display range, and a frequency response flatness of ± 0.5 dB. Sensitivity -98 dBm, with residual responses (no input signal) less than -85 dBm.

Source: https://assist.dla.mil -- Downloaded: 2026l0-22T20:21Z
Check the source to verify that this is the currentversion before use.

TABLE IV. Test harness parts list

DESIGNATION	PART	DESCRIPTION
P1	Connector	TPS plug for RG 195
P2	Connector	TPS plug for RG 58
P3	Connector	TPS plug for RG 58
P4	Connector	ITT Cannon DEM-9S or equivalent
R1	Resistor	50-ohm termination
R2	Resistor	50-ohm termination

NOTE : Wire, T-connectors, RG58 coaxial cable, RG195 coaxial cable, per Figure 5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:21Z
Check the source to verify that this is the current version before use.

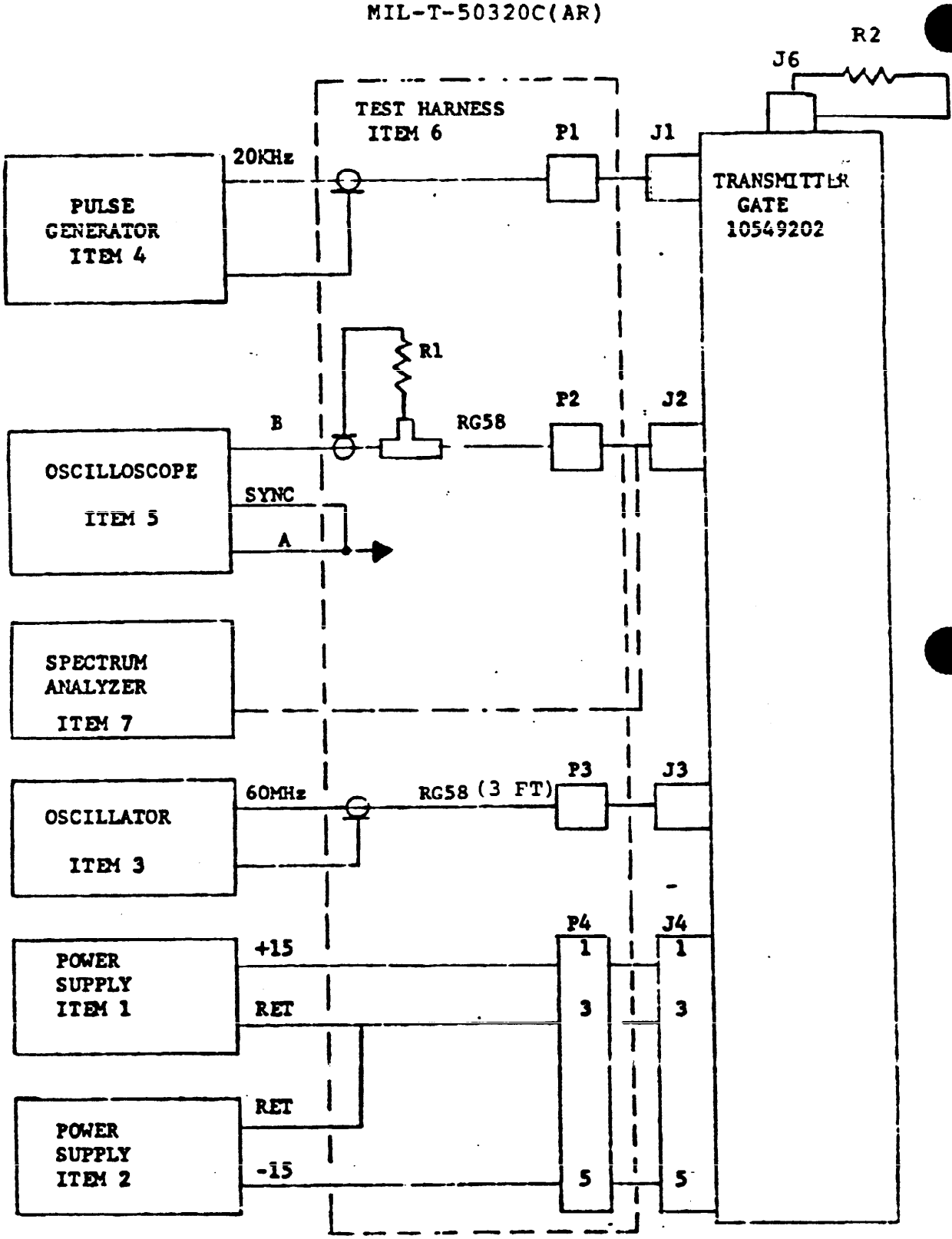


FIGURE 5. Test setup for transmit gate.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:21Z
Check the source to verify that this is the current version before use.

4.7 Methods of inspection.

4.7.1 Environmental. Environmental tests shall be conducted in accordance with the test conditions, general test performance guidance, and individual test methods of MIL-STD-810 except as specified herein. Test facilities, chambers and apparatus used in conducting the tests contained in this specification shall be capable of meeting the conditions required.

4.7.1.1 Vibration. The test shall be performed as specified in Method 514.3, Category 8, Procedure I, o MIL-STD-810 as modified by 3.3.1 herein. In Step 1 of Procedure I only steps 4 through 8 of paragraph 11-2.1 apply. The transmit gate shall be attached to the vibration generator directly or with a fixture, and securely held by its normal means of attachment. After vibration, the transmit shall be tested as specified in 4.7.2 and 4.7.3 herein to determine compliance with 3.3.1.

4.7.1.2 Shock. This test shall be performed as specified in Procedure I, Section II of Method 516.3 of MIL-STD-810 except that:

a. The post test functional check (Step 5 of Procedure I) need not be conducted until completion of all shock impulses.

b. The shock pulses shall be 15 g's ± 10% amplitude and 11 ms ± 10% duration.

c* The maximum g shall occur at 5.5 milliseconds ± 10%.

d. The shock measuring system shall have a flat frequency response within ± 10% from 20 to 2,000 hertz.

After the application of all the shock impulses, the transmit gate shall be tested as specified in 4.7.2 and 4.7.3 to determine compliance with 3.3.2.

4.7.1.3 Altitude (operating and non-operating) This test shall be performed as specified in procedures II and III, "Method 500.2 of MIL-STD-810. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.3.

4.7.1.4 Temperature.

4.7.1.4.1 High (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 501.2 of M1L-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in .4.7.2 and 4.7.3 for compliance with 3.3.4.

Source: https://assist.dla.mil -- Downloaded: 2026l0-22T20:21Z
Check the source to verify that this is the currentversion before use.

4.7.1.4.2 LOW (operating and non-operating). This test shall be performed as specified in procedures I and II, Method 502.2 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.5 Humidity (non-operating) This test shall be performed in accordance with Procedure III, Method 507.2 of MIL-STD-810 except that in Step 7, checkout shall be performed at the conclusion of the test only, with the item at standard ambient conditions.

4.7.2 Electrical. Continuity shall be inspected by standard measuring equipment (SME) and visual examination for compliance with 3.40

4.7.3 Performance. All transmit gates shall be tested as follows:

Connect the test equipment listed in Table III and a transmit gate as shown in Figure 5.

b. Adjust power supplies (Table III, items 1 and 2), oscillator (Table III, item 3), and pulse generator (Table III, item 4) so that the voltages and signals required by 3.5 can be delivered.

c. Disconnect oscillator (Table III, item 3).

d. Use oscilloscope (Table III, item 5) to monitor modulation trigger (Table III, item 4) input signal on channel A, and transmit gate output signal on channel B with modulator trigger used as external sync.

e. Verify that when a modulation trigger pulse train is applied to J1 as described in 3.5.2, the transmit output J2 shall be a D.C. level adjustable as follows:

(1) To within 0 $\pm$ .002V of ground using R24 (transmit gate mod D.C. offset adjustment) as described in 3.5.2.a and Figure 1.

(2) For maximum flatness using R31 (transmit gate D.C. offset adjustment) as described in 3.5.2.b and Figure 2.

(3) Remove the 50 ohm load from J6 and verify that the output is as described in 3.5.2.c. Replace the 50 ohm load after measurement.

f. Verify that the transmit gate output J2 shall be a cosine square RF burst as specified in 3.5.3 and Figure 3 when the oscillator (Table III, item 3) is connected to J3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:21Z
Check the source to verify that this is the current version before use.

g. Disconnect oscilloscope (Table 111, item 5) and connect spectrum analyzer (Table III, item 7) to J2. Verify that 1st and 2nd side lobes of the output spectrum are within the limits specified in 3.5.4 and shown in Figure 4.

h. Verify that with the test conditions of 3.5.3, but with the pulse generator (Table III, item 4) disconnected, the RF signal measured at J2, shall meet the requirements of 3.5.5.

4.8 Soldering. Soldering processes shall be inspected by certified inspectors to the requirements and acceptance criteria of MIL-STD-2000 as specified in paragraph 3.8.

5* PACKAGING

5.1 General. Preservation, packing, and marking shall be in accordance with Packaging Data Sheet PS1054902. The level of protection shall be as specified in the procurement document.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The transmit gate is used In Receiver-Transmitter RT-860, Part of Radar Set AN/vPS-2, and in Test Set Sub-assembly MX-8048, part of Radar Test Set AN/TPM-22. The transmit gate produces a 60-megahertz (MHZ) burst at a 20-kilohertz (KHz) rate. In the test set it is used to check the performance of Radar Set AN/VPS-2.

6.2 Acquisition Requirements. procurement documents shall specify the following:

6.2.1 Procurement requirements.

a. Title, number and date of this specification.
b. Applicable stock number.
c. First Article Sample requirements.
d. Packaging requirements, if other than specified in Section 5.
e. Serialization requirements, if applicable.
f. Certificate of conformation for lot of Shipment or product.

6.3 Submission of contractor inspection equipment designs for approval. Submit two copies of designs as required to: Commander, U.S. Army AMCCOM, ATTN : AMSMC-QAF-I(D), Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:21Z
Check the source to verify that this is the current version before use.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC), may also include drawings prepared by, and identified as, U.S. Army Armament, Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny *Arsenal* drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.5 Subject term (key word) listing.

Fire Control System
Radar Set AN/VPS-2
Radar Test Set AN/TPM-22
Receiver-Transmitter RT-860/VPS-2
Subassembly Test Set MX-8048/TPM-22
Transmit gate
VADS

6.6 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1285-A710)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:21Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-T-50320C (AR)	18 May 1990

3. DOCUMENT TITLE

TRANSMIT GATE: 10549202

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION

c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
	(1) Commercial	(2) AUTOVON
US Army ARDEC Standardization & Specification Office	(201) 724-6674	880-6674

c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:
ATTN: SMCAR-BAC-S/Bldg 6 Picatinny Arsenal, NJ 07806-5000	Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:21Z
Check the source to verify that this is the current version before use.